From Prompt Engineering to RAG

Generative AI Solution Development

Learning Objectives

- Describe prompts and prompt engineering.
- Discuss various techniques and best practices of prompt engineering.
- Describe nuances: RAG, prompt engineering, fine-tuning, and pre-training.
- Identify use cases where RAG can be used to improve the quality, reliability, and accuracy of LLM completions.
- Describe the core components of the RAG architecture.
- Connect Databricks capabilities with the various components of RAG.

databricks

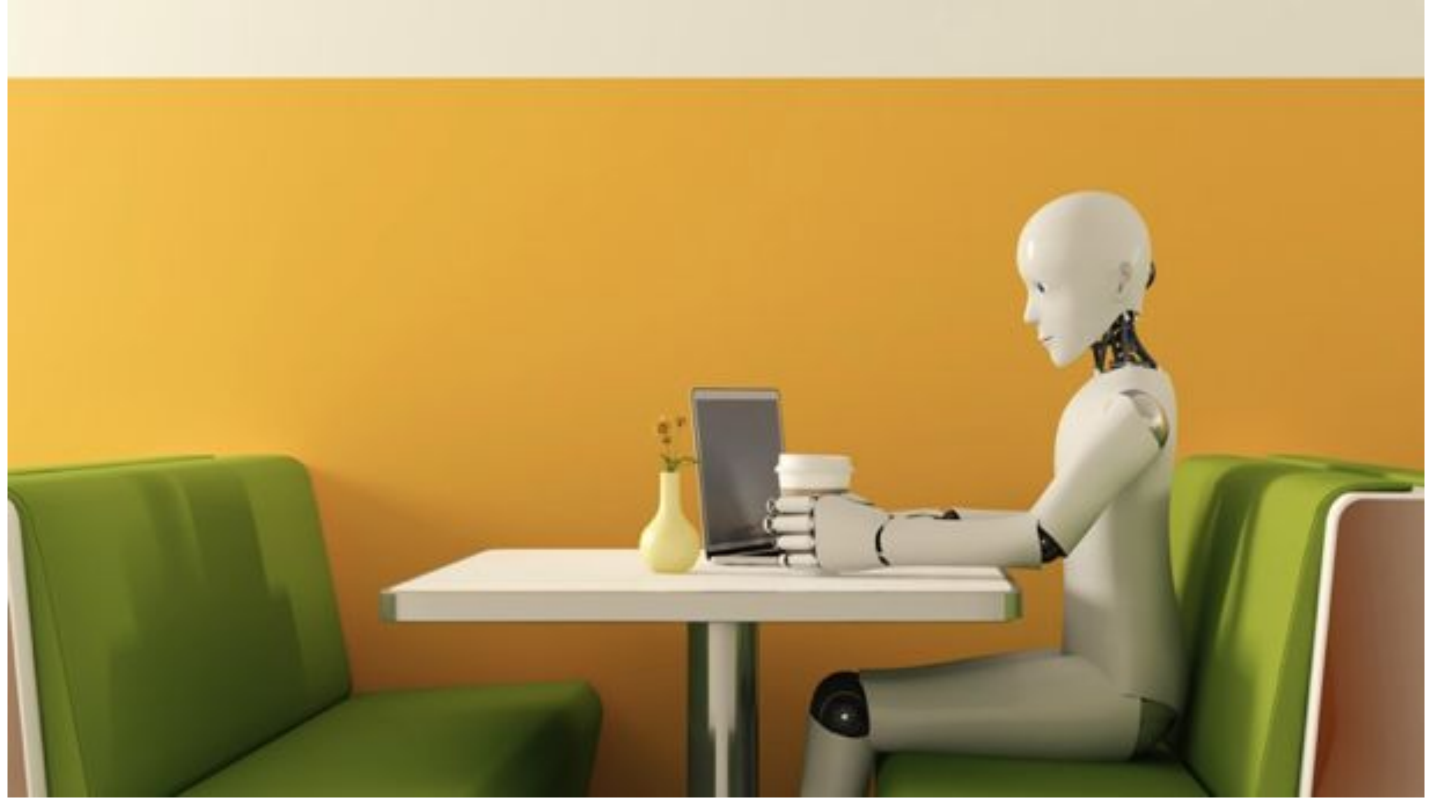
LECTURE

Prompt Engineering Primer

Prompt Basics

Definitions

- **Prompt**: An AI prompt is an **input or query** given to a large language model to elicit a specific response or output.
- **Prompt Engineering**: Prompt engineering is the practice of **designing and refining prompts** to optimize the responses generated by AI models



Prompt Basics

Prompt components

A good prompt usually consists of:

- **Instruction**: A clear directive that specifies what the model should do.
- **Context**: Background/additional information that provides the necessary details to understand the task.
- **Input / question**: The specific query or data that the model needs to process.
- **Output type / format**: The desired structure or style of the response generated by the model.

```
Summarize the following
article and write a list of
top 3 important points in
markdown format that answer
the following question;
“How is climate change
affecting polar bear habitats
and their ability to find
food?”

Article:
<The article discusses the
impact of climate change on
polar bear populations in the
Arctic.>
```

Prompt Engineering Techniques

Zero-shot / Few-shot Prompting

Using or not-using an example

```
Classify the following review into neutral,
negative or positive:

Review: "I absolutely loved this movie! The
storyline was gripping and the acting was
top-notch."
Sentiment: …
```

```
Classify the following movie reviews:
1. Review: "I absolutely loved this movie! The
storyline was gripping and the acting was top-notch."
Sentiment: Positive
2. Review: "The plot was boring and the characters
were unconvincing."
Sentiment: Negative
3. Review: "The movie was a bit slow, but the
performances were excellent."
Sentiment: …
```

Zero-shot Prompting

- Prompt that generates text or perform a task **without providing any examples** or additional training specific to that task.

Few-shot Prompting

- Prompt provides with **a few input-output examples** to guide the model for generating the desired output.

Prompting Chaining

Break tasks into subtasks

- Multiple tasks are linked together, with the output of one prompt serving as the input for the next.
- This method allows for more complex tasks to be broken down into manageable steps.

```
You are a helpful assistant. Your task is to
help answer a question given in a document. The
first step is to extract quotes relevant to the
question from the document, delimited by ####.
Please output the list of quotes using
<quotes></quotes>. Respond with "No relevant
quotes found!" if no relevant quotes were found.
####
Document: "The Amazon rainforest produces 20% of
the world's oxygen..."

Question: "What is the importance of the Amazon
rainforest?"
####
```

```
<quotes>
"The Amazon rainforest produces 20% of the
world's oxygen."
"The rainforest regulates the global climate and
has diverse wildlife."
</quotes>
```

Chain-of-Thought Prompting

"Let's think step-by-step"

- Chain-of-thought (CoT) prompting enhances the reasoning capabilities of LLMs by guiding them to **articulate their thought processes step-by-step**, similar to human reasoning.
- Research in this area has **mixed findings**;
  - Some research showed that simply asking the model to think step-by-step helps to solve reasoning questions.
  - Others showed that models don't follow CoT faithfully (Paper 1, Paper 2, Paper 3).

```
The bakery had 23 bananas. If they used 20 to make a cake and bought 6 more, how many bananas do they have?
```

```
The answer is 27.
```

```
The bakery had 23 bananas. If they used 20 to make a cake and bought 6 more, how many bananas do they have? Let's think step-by-step to solve this.
```

```
The bakery had 23 bananas originally. They used 20 to make the cake. So they had 23-20=3. They bought 6 more bananas, so they have 3+6=9.
The answer is 9.
```

Prompt Engineering
Tips & Tricks

Tip: Prompts are model-specific

A prompt guides the model to complete task(s)

- Different models may require **different prompts**.
- Provide **examples and cues** to guide model's response generation.
- Different **use cases** may require different prompts.
- **Iterative development** is key.
  - Iterate by adjusting the **temperature parameter**—higher for more creative outputs, lower for focused responses.
- Be aware of **bias and hallucination**.

Tip: Format prompts

- Use delimiters to distinguish between instruction and context
  - Pound sign ###
  - Backticks ```
  - Braces / brackets {} / []
  - Dashes ---
- Ask the model to return structured output
  - HTML, json, table, markdown, etc.
- Provide a correct example
  - `"Return the movie name mentioned in the form of a Python dictionary. The output should look like {'Title': 'In and Out'}"`

```
### Instruction ###
Extract the movie title from the
context below and return it in the
form of a JSON object. The output
should look like {"Title": "In and
Out"} and formatted as markdown.

### Context ###
```

Tip: Guide the model for better responses

- Ask the model not to make things up/*hallucinate*
  - `"Do not make things up if you do not know. Say 'I do not have that information'"`

- Ask the model not to assume or probe for sensitive information
  - `"Do not make assumptions based on nationalities"`
  - `"Do not ask the user to provide their SSNs"`

- Ask the model not to rush to a solution
  - Ask it to take more time to "think" → Chain-of-Thought for Reasoning
  - `"Explain how you solve this math problem"`
  - `"Do this step-by-step. Step 1: Summarize into 100 words. Step 2: Translate from English to French..."`

Benefits and Limitations

Benefits

- **Simple and efficient**: The time taken to generate the ideal results is significantly low.
- **Predictable results**: Consistently generate results that meet predefined standards for accuracy.
- **Tailored outputs**: Customization of AI responses to fit specific needs or styles.

Limitations

- The output depends on used model.
- Limited by pre-trained model's internal knowledge. **For external knowledge we need to use RAG**.